

Stage 1 Research Brief

Recovering Structure From Damaged or Poorly Scanned Documents

Field	Detail
Problem ID	ML-T2-022
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Domain	Computer Vision · Document Intelligence
Level	Advanced
Duration	2–3 Weeks

1. Problem Understanding

Historical and institutional archives are often available only as imperfect scans: pages with faded or low-contrast ink, uneven illumination, and physical damage such as torn or missing sections. Simply extracting text from these pages (plain OCR) is not enough, because a system can produce plausible-looking text while silently altering dates, quantities, names, or relationships between fields — and it can silently invent structure in regions where the original content is genuinely unrecoverable.

The real problem this project addresses is therefore twofold: (1) recovering as much genuine structure and text as possible from a degraded page, and (2) being honest about what cannot be recovered, by explicitly flagging uncertain or unreadable regions rather than presenting a guess as fact. A solution that recovers 90% of a page correctly but confidently fabricates the remaining 10% is more dangerous in an archival or legal context than one that recovers 70% and clearly flags the rest as uncertain.

2. Problem Formulation

- Problem statement:** Given a degraded scanned document page, recover its textual content and structural layout while explicitly identifying regions of low confidence or unrecoverable information.
- ML task:** A multi-task problem rather than a single task type: classification (character/OCR recognition, region-type classification, confident-vs-uncertain flagging), regression-style confidence scoring (continuous probability estimates, typically derived from classifier outputs or calibration rather than a standalone regression model), generative sequence modelling (reconstructing missing or torn text using surrounding context), and structured prediction (layout and reading-order detection across the page).
- Input:** Page-level scanned document images, lossless format (PNG/TIFF, not JPEG, to avoid compounding compression artefacts with scan degradation), minimum 300 DPI, accepted in RGB or grayscale colour space.

Output / target: A structured JSON output per page containing: recognised text per region, region-type labels (text / table / image / illegible), a reading-order sequence, and a confidence score for every text span and region. Any span or region below a confidence threshold is explicitly flagged as “uncertain” rather than returned silently as fact.

Unit of prediction: Varies by task head: per character or per word for OCR; per region/bounding box for region classification; per region-pair for reading-order relations; per missing span or token for text reconstruction; and a confidence value attached to every one of the above.

Assumptions:

- Documents are primarily machine-printed (not handwritten); handwritten text recognition (HTR) is a distinct sub-field and is out of scope for this stage.
- English is the primary language for data collection and experimentation. The underlying OCR/layout methods are expected to generalise to other Latin-script languages (e.g. French, German) with retraining or fine-tuning, but this generalisation is not tested in Stage 1.
- Pages have a simple to moderate layout (single or basic multi-column); highly complex forms, ledgers, or dense tables are treated as future extensions.
- Physical damage (torn or missing regions) is localised rather than covering the majority of the page.
- Pages are assumed upright or auto-deskewable by standard preprocessing; skew correction itself is not a primary research focus of this project.

Constraints:

- Zero-cost: only free, open-source software and legally usable public or responsibly generated data.
- No paid APIs, proprietary software, paid datasets, or dedicated/paid cloud hardware.
- The approach must generalise reasonably across varying typography and scan quality, not overfit to one dataset's font or noise profile.

3. Use Case

Primary use case: historical archives (e.g. libraries, government record offices, genealogical archives) that hold large volumes of degraded scanned material and need a trustworthy, semi-automated way to digitise it without silently corrupting the historical record. The same approach is also applicable to legal, healthcare, and corporate record digitisation, where preserving factual accuracy and flagging uncertainty is equally important, though these are not the primary tested context in this stage.

4. Initial Data Strategy

This project focuses on two degradation types, chosen to balance research value against the 2–3 week, zero-cost timeline:

- Faded / low-contrast text — a well-studied, mature problem with strong existing public datasets and tooling.
- Torn / missing regions — a genuinely under-served problem that directly stresses the project's central research question (a model must say “I don't know” rather than hallucinate, since there is no information left to recover).

Skew/orientation correction and substrate/material degradation (stains, water damage, discolouration) are deliberately excluded from the primary scope: skew is a geometric preprocessing problem that does not exercise uncertainty modelling, and substrate degradation is the hardest of the four to simulate or source data for at zero cost. Both are noted as future work.

Data sources

Degradation type	Data source / strategy
Faded / low-contrast text	DIBCO and H-DIBCO benchmark datasets (2013, 2014, 2016) — public, well-established document binarisation datasets built specifically around low-contrast, faint, and unevenly illuminated document images, with ground truth.
Torn / missing regions	No public dataset addresses physical document tears specifically (verified during literature search). Strategy: apply synthetic free-form / irregular masks to clean or DIBCO document images to simulate torn or missing regions, adapting the free-form mask generation algorithm from Yu et al. (gated convolution inpainting) or the stroke-based Quick Draw Irregular Mask Dataset (QD-IMD), both of which produce more realistic, non-rectangular damage shapes than simple geometric cutouts.

Known data-quality concerns to track during implementation: class imbalance between degraded and clean regions, risk of leakage if synthetic masks are too easy to detect versus real damage, and legal/licensing verification for every dataset before use.

5. Existing Solutions

- PreP-OCR (2025) — two-stage pipeline combining image restoration with a ByT5 post-OCR correction model; reports 63.9–70.3% character error rate reduction on degraded historical documents across English, French, and Spanish.
- OCR4all — achieved an 84% error reduction on 19th-century Gothic typefaces through domain-specific fine-tuning, demonstrating that historical OCR is achievable without exotic methods.
- DocRevive (2026) — unified pipeline combining OCR, an occlusion detector, a masked language model (RoBERTa) for missing-word prediction, and diffusion-based inpainting to visually reintegrate reconstructed text. The closest existing system to this project's torn/missing-region sub-problem, though it targets synthetic occlusions (stamps, scribbles, rectangles) rather than irregular physical tears.
- Layout-Aware OCR in Black Digital Archives (2025) — layout-aware OCR pipeline evaluated on real historical newspapers using an unsupervised evaluation framework, relevant because ground-truth transcriptions may not always be available for archival material.

6. Literature / Technical Research

Uncertainty and confidence

Guan et al., “Teaching VLMs to Admit Uncertainty in OCR from Lossy Visual Inputs” (ICLR 2026) — trains a vision-language OCR model with reinforcement learning (GRPO) to tag unreliable spans in its own transcription rather than guessing, reporting an uncertainty-tag F1 of 0.685. Introduces the Blur-OCR benchmark (107,520 training / 2,048 evaluation synthetically degraded images) built specifically for this problem. This is the closest conceptual match to this project's central research question and is treated as an evaluation-layer reference rather than the primary reconstruction model.

Missing / occluded text reconstruction

DocRevive introduces the Occluded Pages Restoration Benchmark (OPRB, 30,078 images) and the Unified Context Similarity Metric (UCSM) for evaluating recovered text. CLII (“Visual-Text Inpainting via Cross-Modal Predictive Interaction”) completes missing image regions and missing text jointly, using each modality to guide the other. A key gap identified during this research: existing work in this space (DocRevive, CLII) addresses synthetic occlusions with regular

or semi-regular shapes (stamps, scribbles, rectangles), not irregular physical tears where the boundary is jagged and characters may be partially cut. This project's use of free-form/stroke-based masks (Section 4) is a direct response to that gap.

Degradation restoration

DIBCO/H-DIBCO-based binarisation work addresses low-contrast, faint, and unevenly illuminated historical text. CAGAN (character attention GAN) restores heavily degraded character patterns using a U-Net-like architecture, representing the GAN/diffusion-based restoration family referenced as Candidate Approach D (Section 7).

Layout and reading order

OmniDocBench provides a joint benchmark and metric set (including Reading Order Edit Distance) for layout detection, OCR, and reading order together. “Reading Order Inference for Complex Document Layouts” (2026) frames reading order as a graph problem over layout entities. These establish the evaluation approach adopted in Section 8.

7. Candidate Approaches

Approach	Why it may work	Potential limitation
A — OCR + confidence thresholding (baseline)	Simple, fast to implement, establishes a reference point for CER/WER before any restoration is attempted.	No genuine reconstruction of missing content; does not address torn regions at all.
B — OCR + masked-language-model reconstruction (DocRevive-style)	Directly addresses reconstruction of missing/degraded text using surrounding context, with a proven approach in the literature.	Existing models (e.g. RoBERTa fine-tuned in DocRevive) are trained on rectangular/regular occlusions and may not generalise well to irregular torn boundaries without retraining on tear-shaped masks.
C — Uncertainty-tagging layer (Blur-OCR-style) on top of B	Directly answers the brief's central question of distinguishing recovered structure from hallucinated structure; aligns output with the project's core research goal.	Reported uncertainty-tag F1 in prior work is 0.685 — a meaningfully imperfect signal that will need calibration and honest reporting of its limits.
D — GAN/diffusion-based visual restoration (CAGAN/DocDiff-style)	Reconstructs the visual appearance of missing text/regions, not just the transcription; useful if a visually repaired output image is a project goal.	Heavier to train within a zero-cost, time-limited setting; risk of producing visually convincing but factually incorrect text, which directly conflicts with the project's uncertainty goal unless tightly paired with confidence tagging from Approach C.

At this stage no single approach is selected as final. The initial methodology (Section 9) treats A as the mandatory baseline and B + C as the primary investigation direction, with D noted as a possible extension if time allows.

8. Evaluation Strategy

- Character Error Rate (CER) and Word Error Rate (WER) — standard edit-distance-based text accuracy metrics, used as the baseline comparison point.
- Expected Calibration Error (ECE) — measures whether reported confidence scores actually match real-world correctness, directly evaluating the uncertainty-quantification half of the project rather than only text accuracy.

- Reading Order Edit Distance (from OmniDocBench) — evaluates layout/structural correctness independent of character-level accuracy, relevant if the layout/reading-order sub-task is included in the final scope.

Per the brief's own emphasis, no single metric (especially not CER alone) will be treated as sufficient evidence of a strong result; text accuracy and uncertainty calibration are reported and discussed together.

9. Initial Methodology

Area	Initial plan
Data	DIBCO/H-DIBCO for faded text; synthetic free-form masks applied to clean/DIBCO images for torn regions (Section 4).
Baseline	Plain OCR (e.g. Tesseract or EasyOCR) with no restoration, measured with CER/WER, as the reference point every other approach must beat.
Candidate models	Approach B (masked-language-model reconstruction) layered with Approach C (uncertainty tagging); Approach D considered as a stretch extension.
Evaluation	CER/WER for text accuracy, ECE for confidence calibration, Reading Order Edit Distance if layout is in final scope.
Experiments	Baseline vs. B vs. B+C, evaluated separately on the faded-text set and the torn-region set, to see whether performance and calibration differ meaningfully by degradation type.
Engineering	Exposed as a simple script/notebook pipeline for Stage 1; a lightweight API or demo interface is a possible later-stage extension, not required now.
Open questions	See Section 10.

10. Open Questions

- Does a masked-language-model reconstruction approach trained/fine-tuned on rectangular occlusions generalise to irregular, stroke-shaped tears, or does it require retraining specifically on tear-shaped masks?
- How realistic do synthetic torn-region masks need to be before results are trustworthy as a proxy for real physical damage?
- Is an uncertainty-tag F1 around 0.685 (as reported in prior work) reliable enough for an archival-quality use case, or does it need further calibration for this project's specific degradation types?
- How should confidence scores be aggregated when multiple task heads (OCR, layout, reconstruction) each produce their own uncertainty for the same region?
- What is a reasonable, defensible confidence threshold for flagging a region as “uncertain” versus accepting it as recovered?

11. References

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